# Fall 2026 DSC 242: Homework 2

Due 11:59PM Friday, October 16

1. (Hoeffding versus Bernstein versus Bennett, 25 points) Let $Z_1, \dots, Z_n \overset{\text{iid}}{\sim} \text{Bernoulli}(p)$ for $p \le 1/2$, and let $t \in (0, 1-p)$. This problem compares the upper bounds for

$$\mathbb{P}\Big[\frac{1}{n}\sum_{i=1}^{n} Z_i \ge t + p\Big]$$

given by Hoeffding's inequality (Lecture 1), Bernstein's inequality (Lecture 3), and Bennett's inequality (Homework 1). Throughout, let $h(x) = (1+x)\log(1+x) - x$ for $x \ge 0$.

   (a) (7 points) Show that the three inequalities give

$$\mathbb{P}\Big[\frac{1}{n}\sum_{i=1}^{n} Z_i \ge t + p\Big] \le e^{-R} \quad \text{with } R \text{ equal to}$$

$$R_{\text{H}} := 2nt^2, \qquad R_{\text{Bern}} := \frac{nt^2}{2(1-p)(p+t/3)}, \qquad R_{\text{Benn}} := \frac{np}{1-p}\, h\Big(\frac{t}{p}\Big),$$

   respectively.

   (b) (6 points) Compute the ratio $R_{\text{Bern}}/R_{\text{H}}$. Supposing $n$ is large, for which $(p,t) = (p(n), t(n))$ does Bernstein's bound improve over Hoeffding's by more than a constant factor in the exponent, i.e., $R_{\text{Bern}}/R_{\text{H}} \to \infty$ as $n \to \infty$?

   (c) (12 points) Show that $R_{\text{Benn}}/R_{\text{Bern}}$ depends on $(p,t)$ only through $x = t/p$, and that it is at least 1 for all $x > 0$, i.e., Bennett's bound is never worse than Bernstein's. Then determine the behavior of this ratio as $x \to 0$ and as $x \to \infty$: for which $(p,t)$ does Bennett's bound improve over Bernstein's by more than a constant factor in the exponent?

   (Hint: For the lower bound, show that $h(x) \ge \frac{x^2}{2(1+x/3)}$ for all $x \ge 0$ by comparing the second derivatives.)

2. (Wainwright 2.18, 25 points) Let $\psi : [0,\infty) \to [0,\infty)$ be a strictly increasing convex function that satisfies $\psi(0) = 0$. The $\psi$-Orlicz norm of a random variable $X$ is defined as

$$\|X\|_\psi = \inf\{K > 0 : \mathbb{E}[\psi(|X|/K)] \le 1\}.$$

   Consider any $\alpha \ge 1$, and $\psi_\alpha(x) = e^{x^\alpha} - 1$. Show that $\|X\|_{\psi_\alpha} < \infty$ if and only if there exist constants $C, c > 0$ such that $\mathbb{P}[|X| \ge t] \le Ce^{-ct^\alpha}$.

3. (BLM 2.27, 25 points) In the setting of Problem 2, suppose that $X_1, \dots, X_n$ are i.i.d. with $\mathbb{E}[X_i] = 0$ and $\|X_i\|_{\psi_\alpha} \le K < \infty$, where $\alpha \in (1,2)$. Prove that there exist constants $C, c > 0$ depending only on $K, \alpha$ such that

$$\mathbb{P}\Big[\frac{1}{n}\sum_{i=1}^{n} X_i \ge t\Big] \le C\exp\big(-cn \cdot \min(t^2, t^\alpha)\big).$$

(Hint: Show $\log \mathbb{E}[e^{\lambda X_i}] \le C\lambda^2$ for $\lambda \in [0,1]$, and $\log \mathbb{E}[e^{\lambda X_i}] \le C\lambda^{\alpha/(\alpha-1)}$ for $\lambda > 1$.)

4. (Vershynin 6.9, 25 points) A random vector $X \in \mathbb{R}^d$ is $\sigma^2$-subgaussian if $v^\top X$ is $\sigma^2$-subgaussian for every (deterministic) unit vector $v \in \mathbb{R}^d$.

   Let $X \in \mathbb{R}^d$ be $\sigma^2$-subgaussian with $\mathbb{E}[X] = 0$, and let $G \in \mathbb{R}^n$ have i.i.d. $\mathcal{N}(0,1)$ coordinates independent of $X$. Show that for any $B \in \mathbb{R}^{n \times d}$ and universal constants $C, C', c > 0$,

$$\mathbb{E}[e^{\lambda^2 \|BX\|_2^2}] \le \mathbb{E}[e^{C\lambda^2\sigma^2\|B^\top G\|_2^2}] \le e^{C'\lambda^2\sigma^2\|B\|_F^2} \quad \text{for all } |\lambda| \le \frac{c}{\sigma\|B\|_{\mathrm{op}}}.$$

   (This is the "sub-Gaussian to Gaussian" comparison used in the proof of the Hanson–Wright inequality in Lecture 5.)

   (Hint: For $G \sim \mathcal{N}(0, I_n)$ and fixed $v \in \mathbb{R}^n$, $\mathbb{E}[e^{v^\top G}] = e^{\|v\|_2^2/2}$. Use this to write $e^{\lambda^2\|BX\|_2^2}$ as an expectation over $G$.)